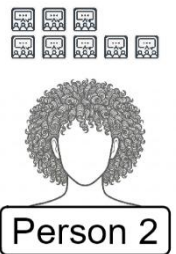


Introduction

Statsville is a quaint fictional hamlet that is nestled in the foothills of the *Regression Mountains* with a beautiful view of *Lake Median*. The **population** is quite small as only **100 residents** between the ages of 13 and 59 live in *Statsville*. The layout of the town is a basic square grid with 20 unique roads – conveniently named with first letters A to T.

The *Mean Diner* is looking to increase business and has decided to start advertising on social media. Before investing into their campaign, management wishes to know what sort of effort is required to advertise on social media. In particular, management needs to know how many social media apps each resident typically uses so they can spend their advertising dollars wisely.

Page 2 includes a diagram of all 100 residents along with an icon  for each social media site (e.g., TikTok, YouTube, Instagram) the resident uses regularly. For example, **Person 2** regularly uses **8 social media sites** as  appears 8 times. Page 3 includes an image of those same residents stratified by their respective age groups (teenagers, twenty-somethings, residents in their 30s, in their forties and fifties). Lastly, page 4 includes an image of each resident grouped by the street in which they live.



We will use this population to explore different sampling techniques to help the *Mean Diner* estimate the average number of social media accounts for residents in *Statsville*.

Teacher Notes & Citations

This activity was heavily influenced by the Random Rectangles activity (Scheaffer, et al., 1996). All images were generated using the Gemini Large Language Model (Google, 2026).

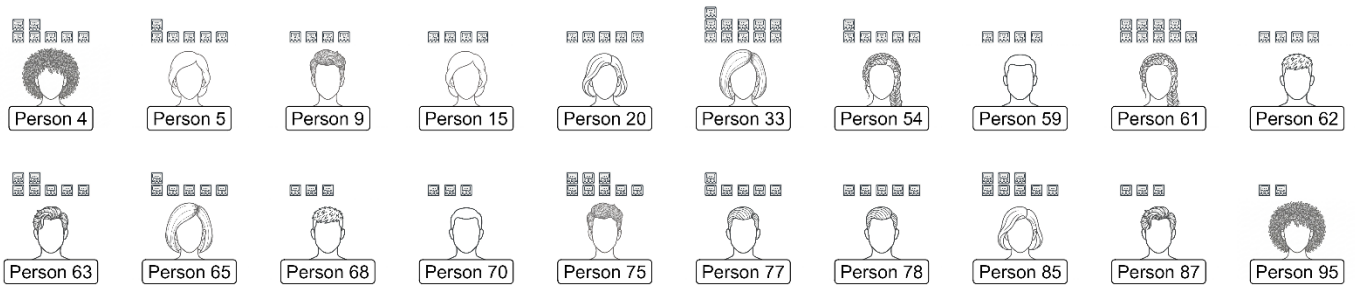
Google. (2026). Gemini 3.6 Flash [Large language model].
<https://gemini.google.com/app>

Scheaffer, R.L., Watkins, A., Gnanadesikan, M., Witmer, J.A. (1996). Random Rectangles. In: Activity-Based Statistics. Textbooks in mathematical sciences. Springer, New York, NY. https://doi.org/10.1007/978-1-4757-3843-8_21

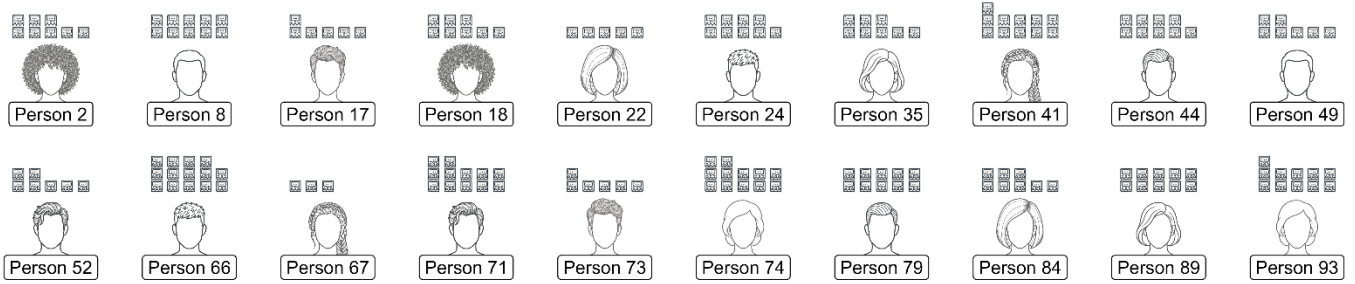
Residents of Statsville

 Person 1	 Person 2	 Person 3	 Person 4	 Person 5	 Person 6	 Person 7	 Person 8	 Person 9	 Person 10
 Person 11	 Person 12	 Person 13	 Person 14	 Person 15	 Person 16	 Person 17	 Person 18	 Person 19	 Person 20
 Person 21	 Person 22	 Person 23	 Person 24	 Person 25	 Person 26	 Person 27	 Person 28	 Person 29	 Person 30
 Person 31	 Person 32	 Person 33	 Person 34	 Person 35	 Person 36	 Person 37	 Person 38	 Person 39	 Person 40
 Person 41	 Person 42	 Person 43	 Person 44	 Person 45	 Person 46	 Person 47	 Person 48	 Person 49	 Person 50
 Person 51	 Person 52	 Person 53	 Person 54	 Person 55	 Person 56	 Person 57	 Person 58	 Person 59	 Person 60
 Person 61	 Person 62	 Person 63	 Person 64	 Person 65	 Person 66	 Person 67	 Person 68	 Person 69	 Person 70
 Person 71	 Person 72	 Person 73	 Person 74	 Person 75	 Person 76	 Person 77	 Person 78	 Person 79	 Person 80
 Person 81	 Person 82	 Person 83	 Person 84	 Person 85	 Person 86	 Person 87	 Person 88	 Person 89	 Person 90
 Person 91	 Person 92	 Person 93	 Person 94	 Person 95	 Person 96	 Person 97	 Person 98	 Person 99	 Person 100

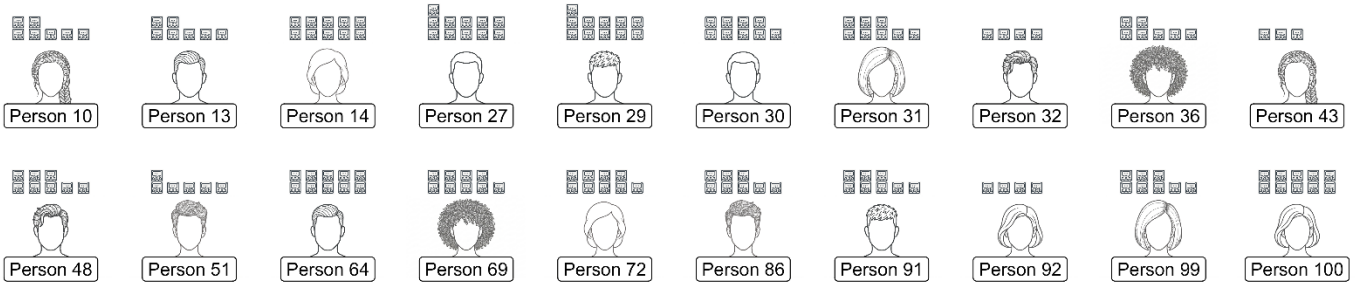
Teenagers



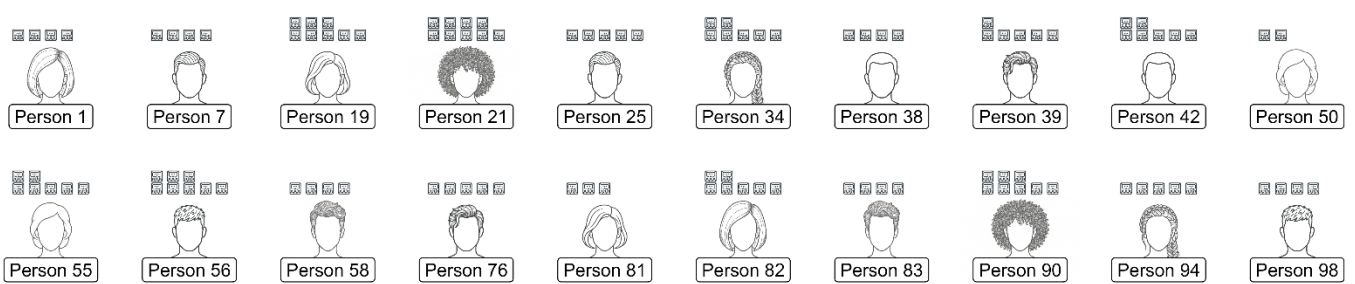
Ages 20-29



Ages 30-39



Ages 40-49



Ages 50-59



Apple Avenue



Beech Street



Chestnut Street



Dogwood Drive



Elm Street



Falcon Avenue



Goetta Way



High Street



Indigo Avenue



Jambalaya Drive



Kookaburra Court



Lemon Lane



Maple Street



Narwhal Alley



Oak Street



Poplar Street



Quinoa Way



Ramblin Road



Sycamore Street



Truffula Road

